

Observing the Real and Imaginary Axes through the Euler Circles (Exercise XXXI)

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Abstract

We observe the real and imaginary axes through the orthogonal Euler circle frame — the radius-1 and radius- i circles of the split-complex numbers — by solving the intersection equations, not by appealing to definitions. The observation yields five machine-checked theorems: the real axis meets the imaginary axis at the single point $(0, 0)$ (solving $\langle a, 0 \rangle = \langle 0, b \rangle$ gives $a = 0$ and $b = 0$); the real axis meets the radius-1 circle at $\{\pm 1\}$ (solving $a^2 = 1$); the real axis does not meet the radius- i circle ($a^2 = -1$ has no real solution); the imaginary axis does not meet the radius-1 circle ($-b^2 = 1$ has no real solution); and the imaginary axis meets the radius- i circle at $\{\pm i\}$ (solving $-b^2 = -1$). The observed conclusion: the two axes intersect at $(0, 0)$, and $i = (0, 1)$ is the intersection of the imaginary axis with the radius- i circle — its unit — not the intersection of the two axes. Zero errors, zero sorry.

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1 The observation

The orthogonal Euler circle frame (Exercises XXVII–XXX) consists of the radius-1 circle $a^2 - b^2 = 1$ and the radius- i circle $a^2 - b^2 = -1$ in the split-complex plane $a + bj$, $j^2 = +1$. This exercise observes the real axis ($\text{im} = 0$) and the imaginary axis ($\text{re} = 0$) through this frame. The method is observational in the strict sense: every result is obtained by solving the relevant equation, not by invoking a definition.

2 The observed table

Observation	Result	Equation solved
real axis $\cap$ imaginary axis	$\{(0, 0)\}$	$\langle a, 0 \rangle = \langle 0, b \rangle \Rightarrow a = 0 \wedge b = 0$
real axis $\cap$ radius-1 circle	$\{\pm 1\}$	$a^2 = 1$
real axis $\cap$ radius- i circle	$\emptyset$	$a^2 = -1$ (no real solution)
imaginary axis $\cap$ radius-1 circle	$\emptyset$	$-b^2 = 1$ (no real solution)
imaginary axis $\cap$ radius- i circle	$\{\pm i\}$	$-b^2 = -1 \Rightarrow b = \pm 1$

The axes meet at $(0, 0)$. A point on both axes has the form $\langle a, 0 \rangle$ (real axis) and $\langle 0, b \rangle$ (imaginary axis); equality forces $a = 0$ and $b = 0$. The intersection is the single point $(0, 0)$.

The axes meet the circles in their units. The real axis meets the radius-1 circle exactly at its unit ± 1 ; the imaginary

axis meets the radius- i circle exactly at its unit $\pm i$. In both cases the axis contributes one equation ($a^2 = 1$, $-b^2 = -1$) and the circle's unit is the solution.

The cross intersections are empty. The real axis does not meet the radius- i circle and the imaginary axis does not meet the radius-1 circle: both equations ($a^2 = -1$, $-b^2 = 1$) have no real solution. The units stay on their own axes.

3 The observed conclusion

The observation answers the positional question directly: the real and imaginary axes intersect at $(0, 0)$ — the point obtained by solving their equations, not by definition. The point $i = (0, 1)$ is the intersection of the imaginary axis with the radius- i circle: it is the unit of that circle, observed on its own axis. It is not the intersection of the two axes. The frame's origin $(0, 0)$, its units $\{\pm 1\}$ on the real axis and $\{\pm i\}$ on the imaginary axis, and the empty cross-intersections form a complete, machine-checked picture of how the axes sit in the Euler circle frame.

4 Honest boundary and position

The content is elementary algebra (linear and quadratic equations over $\mathbb{R}$); the value is observational. The result concerns the frame and its axes; no claim is made about the Riemann zeta function. The frame observes numbers and circles; the analysis of ζ remains the open span of the bridge.

5 Formalization

The proof is a single Lean file (proof.lean, 190 lines), importing only Mathlib.Data.Real.Basic and basic tactics. The split-complex structure (30 lines, as in Exercises XXVII–XXX) is self-contained. Five theorems, compiled with Lean 4.32.2 / mathlib v4.32.2, zero errors, zero sorry.

Theorem	Statement
realAxis_inter_imaginaryAxis	axes meet at $\{(0, 0)\}$
realAxis_inter_unitCircle	real axis meets circle ₁ at $\{\pm 1\}$
realAxis_inter_imaginaryCircle	no real-axis point on circle ₂
imaginaryAxis_inter_unitCircle	no imaginary-axis point on circle ₁
imaginaryAxis_inter_imaginaryCircle	imaginary axis meets circle ₂ at $\{\pm i\}$

6 Conclusion

The observation is complete: the real and imaginary axes, seen through the Euler circles, intersect at $(0, 0)$; their units ± 1 and $\pm i$ lie on their own axes and their own circles; the

cross-intersections are empty. Every entry of the table is the solution of an equation, machine-checked. The axes are observed; the analysis of ζ remains the open span of the bridge.

References